

The Longitudinal Project

Barcelona: pricing and compliance in the short-term rental market

This is the single project running through all twelve sessions. You will not do twelve disconnected exercises. You will build one analysis, and at the end you will defend it in front of the room.

Keep this document. It contains the brief, every deliverable, and the criteria each one is marked against. Nothing in the assessment of this project is hidden from you.

Course: Data Science, Machine Learning and Deep Learning · 12 sessions · ~4 hours of work per week outside class.

What is in here

Part	Contents
Part 1	The brief: the situation, the data, the two questions, the one rule, how you are graded, ethics, and the pitfalls you have been warned about
Part 2	Every milestone M0 to M10, with what to hand in and the criteria it is marked against
Part 3	The other assessed work: mid-course practical, final report, defence, diagnostic test, peer review

1. The situation

Barcelona has the most contested short-term rental market in Europe. In 2024 the city announced it would phase out all licensed tourist apartments by 2028, a policy with no precedent at this scale. Around that decision sit two organisations who want very different things from the same data.

Client A · Direcció de Turisme, Ajuntament de Barcelona (the regulator)

Enforcement capacity is finite: inspectors can visit a few hundred listings a month out of more than fifteen thousand. They want to know which listings are operating **without a valid tourist licence**, so inspections can be targeted rather than random. They also need to defend that targeting publicly. If the model concentrates enforcement in poorer districts, that is a political and legal problem, not merely a technical one.

Client B · a property-management company operating in the city

They manage a growing portfolio and want a **pricing tool**: given a new apartment's characteristics, what should it be listed at? Their current process is a manager eyeballing three listings they consider comparable.

You will work for **both**. They will not always want the same thing, and part of your job is noticing when.

2. The data

Inside Airbnb, Barcelona, snapshot of 24 June 2026. 15,293 listings by 90 columns, vendored in the repository at `data/raw/`.

Inside Airbnb is an activist project that scrapes public Airbnb listing pages to support housing-policy debate. That origin matters. The data was **not** collected for your purpose, has no documentation contract, and contains artifacts of how the scraper works. Part of the work, a real part and not a warm-up, is figuring out what each column actually measures.

***Do not re-download the data.** The snapshot is pinned. Inside Airbnb rotates snapshots quarterly and your results would stop being comparable to everyone else's, including the figures in the course notes.*

What is in it, broadly

Group	Columns
Physical	capacity, bedrooms, beds, bathrooms, property and room type
Geographic	latitude and longitude, 69 neighbourhoods within 10 districts
Host	identifier, tenure, portfolio size
Commercial	price, availability, minimum stay, occupancy and revenue estimates
Reputation	review counts and seven review score dimensions
Regulatory	a license field
Text	listing name, description, host bio, and a JSON list of amenities
Images	photo URLs (optional extension only)

Two warnings, given honestly

- **Some columns are empty or constant.** Not a mistake in the file: the format changed. Find them, and do not assume a column exists because it has a name.
- **Some columns will make your model look excellent.** Be suspicious of good news. If a result surprises you pleasantly, that is a signal to investigate, not to celebrate. This will happen to you more than once, by design.

3. The two questions

Q1 · Regression, for Client B: what should a listing charge per night?

Target: nightly price. You will have to decide what *the nightly price of a listing* actually means in this data before you can model it. That decision is part of the assessment, it is not a formality, and it changes the answer.

Q2 · Classification, for Client A: is this listing operating with a valid licence?

Target: derived by you from the `license` field. The field is free text with several formats and a lot of missing values. Deriving a defensible binary target from it is the first real task, and reasonable people will derive slightly different targets. Document yours.

Optional stretch: predict occupancy. Harder than it looks. A third of listings sit at zero, and the most obviously useful features are contaminated.

4. What you will produce, session by session

After	Milestone	What it is	Graded
S1	M0	Dataset fact sheet and problem statement	formative
S2	M1	Cleaning pipeline, decision log, sealed test set	10%
S3	M2	Preprocessing pipeline and leakage audit	10%
S4	M3	Regression: baseline, linear, regularised	formative
S5	M4	Classification: three models and a threshold recommendation	formative
S6	M5	Validated, tuned model with uncertainty intervals	formative
S7	M6	Ensemble champion and Model Card v1	10%
S8	M7	Neural network from scratch in NumPy, and its PyTorch twin	formative
S9	M8	PyTorch MLP and an honest comparison against M6	formative
S10	M9	Regularisation ablation study	10%
S11	M10	Text and fusion model, and the Final Model Card	formative
S12	-	Report and defence	30%

Formative does not mean optional. Those milestones carry no direct weight, they receive written feedback, and each one feeds the graded milestone that follows it. Skipping one costs you later, not now.

5. The one rule

You will split the data in Session 2, and you will not touch the test set again until Session 12, in class, in front of everyone.

Not « try not to look at it ». Not « avoid using it for tuning ». You will write it to disk, and you will not read that file. Every honest number you report for eleven weeks will come from cross-validation on your training data.

In Session 12 you will state your expected performance out loud, and then the test set is opened. The gap between those two numbers is the most informative result you will produce all semester, and it is only informative if the seal held.

6. How this will be graded

Read this carefully, because it is not how most courses work.

A high score does not earn you a high grade. Raw model performance is capped at roughly 20% of any milestone rubric. The remaining 80% is: was the decision defensible, did you know why you made it, did you check the thing that would have proved you wrong, and can you explain the result to someone who does not know what R^2 is.

A student reporting $R^2 = 0.61$ with a correct protocol and an honest account of its limits will outrank a student reporting 0.87 who cannot explain where it came from. The second student, on this dataset, is almost certainly wrong, and part of what you are learning is how to tell.

You may not always beat a simple baseline by much. That is a real finding, and reporting it clearly is worth more than manufacturing an improvement.

7. Working arrangements

- **Milestones M1 and M2:** assigned pairs. **M3 onward:** re-paired.
- **Final report and defence:** individual. You must be able to defend every choice in your own notebook.
- Discussion across pairs is encouraged. Copying notebooks is not, and you will be asked to explain your code in the defence.
- Budget **about four hours per week** outside class.

8. Ethics, and not as an appendix

This project has a real subject and real consequences, and you are expected to engage with them rather than note them in a closing paragraph.

- **The regulator's model targets people.** A false positive is an inspection of a compliant host, a cost imposed on someone who did nothing wrong. Ask who bears it.
- **Compliance is not uniformly distributed across the city.** When you find a pattern, you will have to decide what it means, and whether a model that exploits it is defensible. Geography carries income, and income carries other things.
- **The pricing model has an aggregate effect.** A tool that helps every operator price optimally does not leave the rental market unchanged. Whether that is your problem is a fair question. Argue it.
- **The data describes identifiable people.** Host names and photos have been removed from the copy you receive. Consider whether the analysis you are doing is one the hosts would recognise as fair.

You will not be graded on reaching a particular ethical conclusion. You will be graded on whether you noticed the questions and reasoned about them.

9. Deliverables at the end

- **Reproducible notebooks** that run top to bottom on a fresh checkout, with fixed seeds and a pinned environment.
- **A report, six pages maximum:** problem, data, method, results with uncertainty, interpretation, limitations, and a recommendation to each client.
- **A Final Model Card:** what it does, how well, for whom it fails, and when it should not be used.
- **A defence:** 12 minutes, 6 minutes of questions, sealed test opened live.

10. Pitfalls, stated in advance

You have been warned about all of these. Being caught by one anyway is normal. Being caught by one *and not noticing* is what costs marks.

Pitfall	
1	Cleaning or exploring before splitting
2	Fitting any transformer on data that includes your validation fold
3	Treating a suspiciously strong feature as good luck
4	Reporting a difference smaller than your fold-to-fold spread
5	Random K-fold on data with obvious group structure
6	Comparing a tuned model against an untuned baseline
7	Imputing a value that cannot exist
8	Dropping rows without checking what they had in common
9	Choosing 0.5 as a threshold because it is the default
10	Presenting feature importance as causation
11	Concluding « deep learning is better » or « deep learning is useless » without a controlled comparison

How the marks are distributed

Instrument	Type	Weight
Session knowledge checks (best 9 of 11)	graded	10%
M1 Cleaning and decision log	graded	10%
M2 Pipeline and leakage audit	graded	10%
A5 Mid-course practical, on a dataset you have never seen	graded	15%
M6 Ensemble champion and Model Card v1	graded	10%
M9 Regularisation ablation	graded	10%
Final report and notebook	graded	20%
Oral defence	graded	10%
Diagnostic reasoning test	graded	5%
M0, M3, M4, M5, M7, M8, M10	formative	0%

Universal rule. In every rubric in this document, raw predictive performance is worth **at most 20%**. Reasoning, protocol correctness and honesty carry the rest.

M0 · Dataset fact sheet

Formative, after Session 1.

Hand in: one notebook containing the shape and dtypes; the columns that are empty or constant; three questions the data can answer; one it cannot, with the reason; and a one-paragraph problem statement naming the client, the unit of observation, the target, and a success criterion a non-technical person could check.

What will be looked at: whether you named the unit of observation precisely, and whether you said how many *independent* units there really are. Those two numbers are not always the same, and noticing that is the point of the exercise.

M1 · Cleaning pipeline and decision log ■ **graded, 10%**

After Session 2.

Hand in: notebook, decision log table, and `test.parquet` written and untouched. The decision log has one row per cleaning decision, with four columns: **what I did** / **why** / **what I would have lost otherwise** / **what this could bias**.

Criterion	Weight	Excellent	Adequate	Poor
Split discipline	20%	Split first, on the correct grouping, justified from measured group structure	Split first, random, grouping not considered	Split after cleaning, or not sealed
Defect identification	20%	Finds all major categories, including deceptive duplicates and empty columns	Finds the obvious ones (types, missing)	Misses type coercion or missingness
Missingness reasoning	25%	Identifies the mechanism per column, measures the bias from dropping, matches imputation to mechanism	Notes the rates and imputes plausibly	<code>dropna()</code> or mean imputation, unexamined

Criterion	Weight	Excellent	Adequate	Poor
Decision log quality	25%	Every decision has a defensible <i>why</i> and a stated risk	Most decisions justified	The log is a list of commands
Reproducibility	10%	Runs clean top to bottom, seeded	Runs with minor fixes	Does not run

Common mistakes: `dropna()` on the whole frame; imputing review scores with the mean when those listings have no reviews; `drop_duplicates()` deleting multi-unit operators; cleaning before splitting and not noticing.

M2 · Preprocessing pipeline and leakage audit ■ *graded, 10%*

After Session 3.

Hand in: a ColumnTransformer and Pipeline object, plus a written leakage audit. The audit names each suspicious column, its mechanism, the **measured** inflation it causes, and the decision you took.

Criterion	Weight	Excellent	Adequate	Poor
Pipeline correctness	25%	All fitted transforms inside the pipeline; nothing fitted on validation data	Pipeline used; one transform fitted outside	Manual preprocessing on the full frame
Traps found	25%	Finds more than the obvious one, and explains the mechanism of each	Finds the most direct one	Finds none; reports the inflated score as a result
Inflation quantified	15%	Reports before and after with a correct protocol	Reports that it « improved »	No measurement
Encoding choices	20%	Cardinality-aware; target encoding handled fold-safely or rejected with a reason	Reasonable one-hot throughout	Encodes 69 categories without comment, or leaks via target encoding
Feature justification	15%	At least three engineered features defended on domain grounds	Features created, thinly justified	Features created with no rationale

Common mistakes: fitting a scaler before the split; target-encoding a high-cardinality column on the full training set; declaring a column a leak without measuring the inflation; assuming the only problem is the column whose name resembles the target.

M3 · Regression

Formative, after Session 4.

Hand in: baseline, then linear, then regularised, with residual plots at each step and a justification of your metric choice.

What will be looked at: did you establish the do-nothing baseline *before* modelling, and did you justify the target transformation rather than copying it?

M4 · Classification

Formative, after Session 5.

Hand in: three models under one protocol, a completed comparison table, and a threshold recommendation with explicit cost assumptions.

What will be looked at: is the threshold derived from a stated cost ratio, or is it 0.5 with a paragraph attached? Is accuracy still being reported as the headline on an imbalanced target?

M5 · Validated and tuned model

Formative, after Session 6. The pivotal one.

Hand in: a tuned model, a cross-validation report annotated with uncertainty intervals, a written defence of your methodology, and any decision you took about **what the target actually means**, with its justification.

What will be looked at, in priority order:

- Did you conclude *the difference is within noise* where the intervals overlap, rather than picking the larger number?

- Did the group structure make it into your cross-validation strategy?
- Is every decision about the target tied to *which client* you are serving?

This is the checkpoint that predicts final-project quality better than any other. It carries no marks and it will receive real written feedback. Treat it accordingly.

M6 · Ensemble champion and Model Card v1 ■ *graded, 10%*

After Session 7.

Hand in: a benchmark notebook and Model Card v1.

Your Model Card v1 must contain:

- the intended use and the client it serves
- performance with uncertainty intervals, and the validation protocol
- top features by permutation importance, **with a stated caveat**
- subgroup performance, including by district
- known failure modes, and the conditions under which it should not be used

Criterion	Weight	Excellent	Adequate	Poor
Mechanism understanding	20%	Explains <i>why</i> bagging and boosting reduce error differently; predictions borne out	Describes both correctly	Treats them as interchangeable « better models »
Protocol correctness	20%	Same protocol across all models; tuning inside validation; early stopping handled correctly	Mostly consistent	Compares tuned against untuned
Honest comparison	20%	Reports where the ensemble did not significantly win, with intervals	Reports differences with some uncertainty	Ranks by point estimate
Interpretation	20%	Permutation importance and partial dependence correctly read; states what they do not prove	Produces and describes them	Presents importance as causation
Subgroup and fairness	20%	Finds and quantifies any subgroup gap, and reasons about the enforcement consequences	Notes that a gap exists	No subgroup analysis

***Common mistakes:** treating a sparse-versus-dense `TypeError` as a blocker rather than reading it; reporting impurity importance instead of permutation importance; asserting that a random forest cannot overfit; benchmarking on a single split.*

M7 · Network from scratch

Formative, after Session 8.

Hand in: the NumPy network and its PyTorch twin, with **matching outputs**, plus your completed hand-computation worksheet.

What will be looked at: does your hand-computed gradient match autograd? If it does not, that is the most valuable debugging conversation of the block, and you will be expected to run it yourself.

M8 · PyTorch MLP and honest comparison

Formative, after Session 9.

Hand in: a working pipeline, a gallery of training curves each with a diagnosis, and a documented comparison against your M6 champion under the same protocol.

What will be looked at: is the comparison genuinely like-for-like? And if the classical model wins, do you *explain* that result or apologise for it?

M9 · Regularisation ablation ■ *graded, 10%*

After Session 10.

Hand in: an ablation table covering baseline, early stopping, weight decay, dropout, batch normalisation and the combination, with **three seeds each**, mean and spread, plus a written recommendation.

Criterion	Weight	Excellent	Adequate	Poor
Experimental control	30%	One factor at a time; everything else fixed; seeds reported	Mostly controlled	Multiple changes per row
Seed discipline	20%	Three seeds, spread reported <i>and used in the conclusions</i>	Multiple seeds, spread reported	A single run per configuration
Diagnosis quality	20%	Each configuration's curve correctly classified	Most classified	Curves not examined
Honest conclusions	20%	States where an intervention had no effect beyond noise	Some hedging	Declares every change an improvement
Mechanism explanation	10%	Each remedy tied to the problem it addresses	Remedies described	A recipe list

Common mistakes: one seed; concluding that $+0.004$ is an improvement; forgetting `model.eval()` and mis-reading the result; stacking everything and reporting only the combined row.

M10 · Text, fusion and the Final Model Card

Formative, after Session 11.

Hand in: a bag-of-words baseline, then an encoder-based model, then a fusion model; a master comparison table across all eleven weeks; and your Final Model Card.

What will be looked at: was the simple text baseline built *first*? Did you check whether the text introduces any leakage of its own, and handle what you found? And does the master table make an argument, or merely list numbers?

A5 · Mid-course practical ■ *graded, 15%*

Take-home, one week, after Session 6.

A **five-hour budget**, on a small dataset you have never seen: a different domain, a few thousand rows, supplied by the instructor.

This is the only instrument that tests transfer. Everything else you are assessed on is measured on a dataset you have lived with for weeks. This one is not, and that is the point.

Hand in: one notebook containing problem framing, the split, exploratory analysis, preprocessing, a baseline, at least two models, validation with uncertainty, and a recommendation.

Criterion	Weight
Workflow completeness and correct ordering	25%
Split and validation strategy appropriate to <i>this</i> data's structure	20%
Leakage identified and handled	20%
Metric choice justified for the stated context	15%
Uncertainty reported and used in the conclusions	10%
Performance relative to a sensible baseline	10%

Common mistakes: importing the Barcelona workflow wholesale without checking whether this dataset has the same structure; missing a leakage trap because it works by a different mechanism; no baseline.

Final report and notebook ■ *graded, 20%*

Hand in: reproducible notebooks and a report of **six pages maximum**.

Criterion	Weight	Notes
Problem framing and target definition	15%	Including any decision about what the target means, and which client it serves
Methodological correctness	25%	Split integrity, validation, leakage, protocol consistency
Results with uncertainty	15%	Intervals, not point estimates
Model comparison across all three blocks	15%	Classical against deep learning, argued from your own evidence
Interpretation and limitations	15%	Including subgroup performance and ethical reasoning
Communication to a non-technical client	10%	Client A and Client B need different documents
Reproducibility	5%	Fresh-checkout run, seeds, pinned environment

Automatic cap at C if the test set was used before Session 12, or if a reported number cannot be reproduced by running your notebook.

Oral defence ■ *graded, 10%*

Twelve minutes of presentation and six minutes of questioning, in Session 12. **Before the sealed test set is opened, you state your expected performance.**

Criterion	Weight
Defends specific decisions under questioning	30%

Criterion	Weight
Accuracy of the pre-reveal prediction, and the quality of the explanation for any gap	20%
Explains limitations without being prompted	20%
Communicates to the stated client rather than to the instructor	15%
Handles a challenge to a choice made, revising or defending with reasons	15%

On that 20%: you are rewarded for a well-explained gap, not only for a small one. A student who predicted 0.62, got 0.57, and correctly attributes it to fold variance and a shift between hosts scores higher than one who predicted 0.62, got 0.62, and cannot say why.

Diagnostic reasoning test ■ *graded, 5%*

Closed book, no coding from scratch. You will be given an analysis you have never seen and asked whether it can be trusted.

Part	Task	Marks
1	Find the errors: a supplied notebook containing methodological defects. Name each, explain the consequence, state the fix	20
2	Ship which model? Two models with full metrics plus a business constraint. Choose and justify	10
3	Read the curve: an unlabelled learning curve. Diagnose and prescribe	10
4	Short conceptual items spanning all three blocks	10

Peer review

At each graded milestone you review one peer's deliverable against the rubric, about thirty minutes. Reviews are submitted and count toward the knowledge-check component. Applying a rubric to someone else's work is the cheapest way to internalise it, and it tends to surface the assumption that a higher score is the goal.

If you remember one thing from this document. You are not being marked on the size of your numbers. You are being marked on whether you can be trusted with them.